

Quality Requirement	Architectural Decision	Justification
Scalability	* Amazon Kinesis Data Streams for high-throughput horizontal ingestion scaling. * Concurrent auto-scaling of serverless AWS Lambdas. * TimescaleDB Hypertables (raw_telemetry, clean_telemetry, vehicle_events) partitioned automatically by time intervals.	* Kinesis acts as an elastic buffer that absorbs high-frequency telemetry spikes. * AWS Lambda naturally scales concurrent execution counts up and down to match ingestion volume. * TimescaleDB hypertables prevent performance degradation associated with massive B-Tree index sizes by isolating inserts into specific time-based chunks, maintaining fast, linear write speeds as active fleet data scales
Reliability & Fault Tolerance	* AWS Lambda SQS Dead-Letter Queue integration coupled with a database telemetry_errors capture layer. * Flyway Chronological Migrations & Split-Directory Reverse Scripting (paired V and U scripts)	* Isolates malformed JSON payloads at the compute layer, while database processing exceptions capture raw payloads natively alongside a synced_to_cloudwatch tracking mechanism to eliminate data loss. * Flyway schema tables enforce an immutable structural timeline. Split up/down files allow automated verification and deterministic rollback execution (undo) without corrupting production PostGIS or time-series partitions
Performance	* TimescaleDB Continuous Aggregates utilizing background refresh policies * PostGIS Native Spatial Indexing (GIST) paired with C-optimized geometry functions (ST_ClusterDBSCAN, ST_Contains, ST_Intersects).	* Asynchronously pre-computes operational metrics outside the ingestion pipeline, keeping read workflows ultra-fast without loading disk memory during writes.

	* Partial Unique Indexing for state tracking. * Keyset Pagination (p_before_start_time) for data retrieval. * Sliding-window Telemetry Snapping Views	* Offloads heavy GIS math, geofence evaluations, and stop/event clustering to native spatial database indexes (R-Tree), entirely removing slow, resource-heavy in-memory processing loops from application code. * Partial indexes target exclusively active ('open') trip rows, keeping operational indexes exceptionally small and O(1) efficient. * Keyset pagination bypasses costly standard SQL OFFSET scans, guaranteeing identical query speeds for deep page requests regardless of total table growth. * Binds real-time fleet asset tracking lookups to a time window, preserving system performance during active map interaction.
Security & Privacy	* AWS Cognito User Pools for secure JWT generation and Role-Based Access Control (RBAC). * TLS 1.2 enforcement and strict rate-limiting traffic policies configured on AWS API Gateway.	* Offloads user sign-in, session tokens, and role evaluation to a dedicated provider. Token verification happens at the API Gateway edge via native authorizers, dropping unauthenticated or rate-limited requests before they ever consume downstream database or compute resources..
Observability	* AWS CloudWatch custom dashboards and logging groups attached to Lambda handlers, integrated directly with an explicit telemetry_errors.synced_to_cloudwatch view flag.	* Provides end-to-end trace visibility. Ingestion schema parse failures retain their full raw payload inside the database while toggling an observability flag, allowing cloud log routers to surface pipeline edge cases without destabilizing the application.
Maintainability	* Infrastructure as Code (IaC) implementation via AWS CDK for	* AWS CDK ensures repeatable infrastructure definitions and minimizes manual deployment

	pipeline deployments, paired with local Docker Compose replication. * Database schema version control managed through Flyway split-directory structures (/migrations and /reverse_migrations).	variations. Docker Compose mirrors target services (PostgreSQL/TimescaleDB/PostGIS) locally for instant verification. * Organizing Flyway configurations to process distinct up (V) and down (U) directories eradicates environment drift while safely decoupling ongoing architecture modifications from recovery rollbacks .
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